

Project 8: Final Group Project

CS6230 AI Tools

This term, we emphasized a few themes:

- (1) variety of modeling problems (e.g., regression, classification, time series), associated techniques, and how to evaluate predictions
- (2) descriptive versus predictive models
- (3) engineering features to maximize tractability by machine learning models
- (4) improving models through the inclusion of complementary data

In your final project, you will apply what you've learned to a data set of your choosing. You will present your work in a 15-minute presentation. You are encouraged to work in teams of 2-3 people.

Project Proposal

Your team must write a project proposal, and have it approved by the instructor. Find three data sets you propose to work on. For each data set, you must:

1. Define an appropriate machine learning problem (e.g., regression or classification) you will attempt to solve. In this definition, you must define the output variable, examples of metrics you would use to evaluate solutions, and a quick description of criteria for a "good" solution (e.g., $\geq 80\%$ accuracy).
2. Given the background for understanding this problem, the utility / applications, and potential stakeholders.
3. Describe what modeling techniques you intend to use and ideas for feature engineering.
4. List 2 examples of complementary data sets you can use to achieve better models. Data sets such as zip code $\rightarrow$ GPS coordinates or census data such as median household income, educational achievement, etc. for each zip code are all perfectly acceptable. You do not need to have these data sets in hand for the proposal, but you will be required to use them as part of the final project.
5. Demonstrate that you can access the data set by including a screenshot showing the first 10 records.

To avoid duplicate topics, I will pick from the three data sets and tell you which data set to use for your project.

Data Sets

You should find a "rich" data set and associated problem. Examples include:

[Predict the Scores of Reddit Comments](#)

[Forecast the Number of Taxi Rides Per Day](#)

Forecast Grocery Store Sales Per Day

Project Requirements

For the project, you must:

1. Create a baseline model using the chosen data set with no external data sets, minimal feature engineering, and a linear model.
2. Apply the following techniques to improve your model:

Feature Engineering: You can try engineering other representations of the features that may give better signals. For example, you could try:

Add More Data: Most gains from model performance come from including additional variables from complementary data sets. **You will be required to use two additional data sets to complement the main data set.** One example of a complementary data set would be the American Community Survey from the U.S. Census. The ACS gives median household income, unemployment rates, age distributions, and more by zip code. You will need to find one or more complementary data sets that you can join with the original to add more variables. Once you acquire, clean, and select the desired columns from the ACS data set, you can join it with your current data set. Other data sets might include population densities, crime rates, or school ratings.

More Complex Model Types: Try other model types such as those based on decision trees / random forests, support vector machines with non-linear kernels (e.g., RBF), and simple feedforward neural networks. Note that you will need to tune the hyper-parameters (e.g., C and epsilon for SV Regression).

3. Summarize and explain the improvement from the baseline model to your best model. What features, data, and techniques worked best? What insights into the data and techniques did you gain from your efforts?
4. Create a minimum model for explanatory purposes. You should perform feature selection to identify a model that offers similar prediction performance to your best model but a smaller set of features. Which features are included in this model? Create visualizations to explain the decision rules or boundaries used by this model.
5. Evaluate the applicability of your model. Is your model performing sufficiently to be trusted and meet the needs of the stakeholders? What insights into the data or problem does your explanatory model provide?

Presentation Requirements

You will create and deliver a 15-minute PowerPoint presentation on your project during finals week. Be generous with visuals. Try to tell an interesting story. Your presentation must cover:

- An overview of the problem and data set you chose. Explain who the stakeholders are and the potential applications / importance. Describe your criteria for success.
- A description of your baseline model and its performance

- A review of approaches you tried to achieve your best possible model. What worked? What didn't? What did you learn about the data and the techniques through this process?
- A description of your minimal explanatory model.
- Key takeaways that are relevant to your colleagues and/or stakeholders.

Submission

You will need to submit three separate things to Canvas:

- Project proposal
- Your presentation and supporting files (e.g., notebooks, source code)